

1 Quick Review

Even more important than the definition of a vector space, is how we categorize and work with them. This week you will learn just how we do that and even how we move from one vector space to another!

In every math class you have probably discussed the notion of a *coordinate*. But now the question is: *What is a coordinate?* Vectors have coordinates, so it cannot just be a point in space since vectors are not restricted to being just arrows in space.

So, we have to find if there is a systematic way to describe a vector's qualities for *every* vector in a vector space. Recall that the most important properties of a vector space are:

1. Closure under vector addition
2. Closure under scalar multiplication

Using these two properties we can extrapolate that there must be some set of vectors that spans the entirety of the vector space. We call this set of vectors a **basis**. Since every vector v can be described using a basis, then we can associate a quantity of each vector in a basis (a basis vector) all adding together to form v : a linear combination!

So we did it, coordinates are just a linear combination of basis vectors!

Now we can ask some questions about bases:

- What does the number of vectors in a basis space mean?
- How can I get from basis to another basis?
- Can a basis have an amount of vectors that I'd like?
- Is every vector space a basis?

As we work with bases, the answers to these questions should become quite obvious!

Definition 1: Basis

Let V be a vector space. Let $S \in V$ be a finite set of vectors. S is a **basis** for V if:

1. S spans V .
2. S is linearly independent.

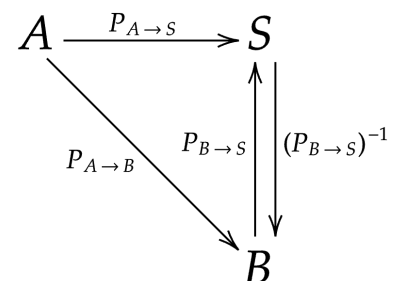
Definition 2: Dimension

Denoted by $\dim(V)$, the dimension is the number of vectors required to fully-define a vector in a vector space. This is also the amount of basis vectors in a vector space's basis.

When changing bases, it is often very helpful to visualize the process with a diagram as shown on the right. In this example

$$P_{A \rightarrow B} = (P_{A \rightarrow S})(P_{B \rightarrow S})^{-1}.$$

This visually demonstrates how we have to move from one basis to another. As you can see to go from A to B , we have to go through the standard basis S .



2 Problems

1. A set of objects $S = \{(1, 3, 5), (6, 7, 2), (2, 0, 1)\}$ is a basis for $\mathbb{R}^3$.

- (a) Is S a vector space? _____
- (b) Is S linearly independent? _____
- (c) If S is linearly independent, prove that it is.

- (d) Does S span $\mathbb{R}^3$? _____
- (e) If S spans $\mathbb{R}^3$, prove that it does.

2. You are give a set of vectors $S = \{(3x^3 + 4x), (5x^2 + 3), (3x^3 + x), (x^4 + 5x^2)\} \in \mathbb{P}_4$.

- (a) Is it possible for S to span $\mathbb{P}_4$? If not, why not?

(b) Is S a linearly dependent set of vectors? _____

(c) Based on your answers for parts (a) and (b), is S a basis for $\mathbb{P}_4$? Why?

(d) You should have found that S is *not* a basis. Do something to make S a valid basis:

(e) How did your answer to part (e) affect the dimension of S ?

3. **(PROOF).** Prove that if a set of vectors S is a basis for some vector space V , then S itself is a vector space.

4. Below are three different bases for $\mathbb{R}^3$.

$$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T = \{(3, 0, 7), (2, 1, 0), (0, 5, 1)\}$$

$$U = \{(3, 0, 0), (0, 2, 6), (4, 5, 0)\}$$

(a) What is the name of S ? _____

(b) Write the transformation matrix for $S \rightarrow T$.

(c) Write the transformation matrix for $S \rightarrow U$.

(d) Write the transformation matrix for $T \rightarrow U$. (Hint: Use the “Ultimate Transformation Triangle!”)

(e) Let $v = (12, 8, -6)$ be defined in terms of the standard basis of $\mathbb{R}^3$. Find $[v]_T$ and $[v]_U$.

5. **(BONUS)**. I promised you guys some applications for linear algebra, so here we go! The information below has been found with the help of <https://courses.cit.cornell.edu/ece4060/Lectures/handout8.pdf> and [Ch 2: What are kets and wavefunctions? | Maths of Quantum Mechanics](#).

Quantum mechanics tells us that some quantities such as the energy $\mathbf{E}$ have a discrete set of values that form the overall quantum state. We can represent this state using an energy basis (a special set of vectors called an *eigenbasis*) $E = \{E_1, E_2, \dots, E_n\}$.

- (a) Write the quantum state ψ_s using the energy basis E with c_i being the coefficients. (Hint: use summation notation!)

- (b) Now imagine we had a *continuous* and *infinite* set of basis vectors associated with some quantity $\mathbf{X}$ (recall that infinite-dimensional bases are possible because vector spaces like $\mathbb{R}^\infty$ exist!). We should also be able to represent the quantum state by forming a linear combination of the continuous and infinite basis $X = \{x_1(t), x_2(t), x_3(t), \dots\}$.

Note that X is a set of functions, which can be vectors.

What vector space might the basis X be for? (Such as $\mathbb{R}^7$ or $\mathbb{P}_3$)

- (c) Since X is a set of continuous and infinite functions, how may we denote the quantum state ψ_s now? (Hint: how do we represent the continuous version of summation?)